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Your ref. 12710797.7/1453
Our ref. KB/P102705EP00

The Hague,
13 October 2016

Re: European patent application No. 12710797.7
Tel HaShomer Medical Research Infrastructure and Services Ltd.

In response to your communication dated 5 April 2016, I herewith file amended claims. For your convenience, both a version showing amendments highlighted and a clean version are enclosed. In addition, I observe the following.

5 Amendments:

Claims 1, 2, 5 and 10-15 correspond to previous claims 6-11 and 13-15, respectively.

Claim 1 finds basis in claims 6 and 12 as originally filed (and as published in WO 2012/123938).

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Claim 2 finds basis in claims 7 and 12 as originally filed.

Claim 3 finds basis in claims 1, 12 and 34 as originally filed. Claim 34 depends on claim 12, which claim refers to (amongst others) claims 6 and 7.

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Claim 4 finds basis in claims 35 as originally filed and further on page 17, lines 4-8 of the application as originally filed.

Claim 5 finds basis in claims 8 and 12 as originally filed.

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Claim 6 finds basis in claim 40 as originally filed.

Claim 7 finds basis in claim 37 as originally filed, as well as on page 18, lines 22-23 of the application as originally filed.

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Claim 8 finds basis in claim 38 as originally filed, as well as on page 18, line 24 of the application as originally filed.

Claim 9 finds basis in claim 39 as originally filed, as well as on page 18, line 25, of the application as originally filed.

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Claim 10 finds basis in claims 9 and 12 as originally filed.

Claim 11 finds basis in claims 10 and 12 as originally filed.

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Claim 12 finds basis in claims 11 and 13 as originally filed.

Claims 13-15 find basis in claims 14-16 as originally filed, respectively.

Previous claims 1-5 and 12 have been deleted. Accordingly, the objections against claims 1-5 based on clarity and lack of support raised in the international search report are no longer relevant.

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Novelty:

Claims 1-15 are novel over WO 2007/022474 (D1), because they are directed to a compound for use in multiple sclerosis. In contrast, D1 is directed to a composition comprising a nucleic acid and a quinolone analog of formulas (I)-(V) for use in the treatment of psoriasis.

Inventive Step:

Claims 1-15 also involve an inventive step in view of WO 2010/113096 (D2) and the publication by Drygin Denis (D3) for the following reasons.

According to the written opinion of the IPRP, D2 would represent the most relevant state of the art. D2 would disclose that compounds which decrease the expression level of at least one gene involved in the RNA polymerase I pathway are efficient drugs in the treatment of multiple sclerosis (MS).

Further, according to the written opinion, D3 would provide an alternative anti MS drug, because D3 would describe analog CX-5461 as a potent and selective inhibitor of RNA polymerase I. Accordingly, the combination of D2 and D3 would make it obvious for a skilled person to try CX-5461 for treating MS.

It is disagreed with the IPRP that the skilled person would combine the teachings of D2 and D3 to arrive at the present invention.

First of all, when considering D2, the skilled person would have no reason to consult D3 for alternative MS drugs, as D3 does not relate to the treatment of MS.

Second, even when consulting D3, the skilled person would not conclude based on D3 that analog CX-5461 would be a potent alternative MS drug, as explained below.

If anything, D3 teaches away of such use. D3 explicitly teaches that CX-5461 specifically affects cancerous cells (transformed cells) while having no effect on non-transformed cells, as explained in the description of Figure 4 on page 1424 of D3, which states that "*CX-5461 exhibits broad anti-proliferative potency in a panel of cancer cell*
5 *lines in p53-independent manner, but has minimal effect on viability of non-transformed human cells*" and in D3, page 1429, first column, which states "...CX-5461 takes advantage of the equivalent qualities of cancer cells and elicits potent and selective antitumor activity, whereas having a minimal effect on non-transformed cells." D3 also discloses that "*CX-5461 was well tolerated at all tested schedules as judged by the*
10 *absence of significant changes in animal body weight*" (see page 1425, second column, last paragraph, therein).

As an autoimmune disease in general, and MS in particular, does not involve transformed cells, the skilled person starting from D2 and looking for alternative MS
15 drugs, would be deterred by D3 from using CX-5461 in order to affect non-transformed cells.

This is in sharp contrast to the instant application which provides unprecedented evidence that CX-5461 can reduce the viability of non-transformed immune cells from
20 both healthy and MS subjects (see, for example, Examples 2-4, Figures 5, 6 and 7A-B of the application as filed).

Furthermore, the skilled person would not expect based on the mechanism of action of analog CX-5461 described in D3 that it would also be suitable as an MS drug. The
25 mere reason that CX-5461 is able to disrupt the binding of the SL1 transcription factor to the rDNA promoter does not automatically make it a suitable as an MS drug, as explained below.

D3 discloses that the mechanism of action of CX-5461 is disruption of binding of the SL1 transcription factor to the rDNA promoter. Such disruption inhibits initiation of rRNA synthesis by the Pol I multi-protein complex (see D3, page 1427, first column, first paragraph).

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In contrast, D2 does not refer to activity of an RNA polymerase I enzyme itself, but rather to the expression of a specifically defined network of genes involved in the RNA polymerase I pathway (see D2, Table 3, pages 120-121). Specifically, D2 discloses that an efficient drug molecule for the treatment of RRMS is a "*drug molecule which*
10 *downregulates a level of expression of at least one gene involved in the RNA polymerase I pathway*" (see, e.g., claim 9 in D2).

However, SL1 (*i.e.* the gene targeted by CX-5461 of D3) is not one of the genes defined as a "gene involved in the RNA polymerase I pathway" according to D2. None of the
15 genes mentioned in D2 that are considered relevant with respect to MS include SL1.

Therefore, based on the mechanism of CX-5461 described in D3, the skilled person would not expect that CX-5461 would have an activity similar to the activity described for the compounds of D2.

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Furthermore, D2 teaches the use of **expression** inhibitors, whereas D3 teaches that CX-5461 selectively inhibits RNA polymerase I **activity** (*i.e.*, in the synthesis of rRNA) while not affecting the levels of mRNA and protein synthesis. Thus, CX-5461 cannot be considered as an **expression** inhibitor based on the teachings of D3, which describes
25 the compound as "*a potent and selective inhibitor of Pol I mediated rRNA synthesis in cancer cells that does not inhibit DNA, mRNA or protein synthesis*" (see D3, page 4118, second column, second paragraph).

Therefore, the skilled person starting from D2 and looking for MS drug alternatives, would not expect that the activity inhibitor described in D3 would have the same activity on MS as the expression inhibitor described in D2, especially considering that D3 explicitly mentions that the activity inhibitor of D3 do not inhibit mRNA expression or protein synthesis.

Summarizing, the skilled person would not arrive at the present invention based on D2 and D3. First of all, the skilled person would not consider consulting D3 when trying to find alternative MS drugs from those described in D2. Furthermore, based on the mechanism of action, the skilled person would not expect that compound CX-5461 would also work as a potent MS drug.

Therefore, the present claims are inventive in view of D2 and D3.

Based on the above, I believe that the present application meets all the requirements of the EPC. In case any objections remain, it is requested that you issue another communication. In case the Examining Division intends to refuse the present application, Oral Proceedings in accordance with Article 116 EPC are requested.

The professional representative,



K.M.L. Bijvank

Enclosures:

Amended claims (clean and marked-up version)